1 Lattices

A lattice is a free $\mathbb{Z}$-module of finite rank with a symmetric bilinear form. Its dual is defined in Section 3.

2 Gram matrices

For a basis $e_1, \ldots, e_n$ the Gram matrix is $G = (b(e_i, e_j))_{i,j}$.

3 Dual lattices

The dual lattice is $L^\vee = \{x \in L \otimes \mathbb{Q} : b(x, L) \subseteq \mathbb{Z}\}$.